Does Increased Cognitive Load Further Effect Lower-Extremity Landing Mechanics During Unanticipated Change of Direction Tasks?

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Introduction: Multi-directional sport requires rapid adjustments to a continuously evolving environment, where athletes must identify, process, and react to ever changing stimuli while also remembering critical information relevant to performance outcomes. Anterior Cruciate Ligament (ACL) screenings attempt to recreate sport-specific environments, where cues to change, direction are reduced to a single arrow or light [1, 2, 3]. However, the ecologic validity of such paradigms has been questioned [4], as this reductionist approach may neglect the additional cognitive demands, which have been shown to influence ACL injury risk factors [5]. While reducing available planning time, through unanticipated change of direction tasks increases ACL risk factors, further increasing visual discriminatory complexity may further challenge athletes while improving ecologic validity. Therefore, the purpose of this study was to determine the effects of increasing cognitive load, through discriminatory complexity, on ACL injury risk factors during unanticipated change of direction tasks. The incongruent flanker task (Fig 1C), which previously has been shown to influence ACL injury risk factors compared to planned conditions [6], was compared to a standard arrow task (Fig 1A) to determine whether more complex unanticipated stimuli further affect lower extremity biomechanics during change of direction (CoD) tasks. We hypothesized that the incongruent flanker task condition will result in increased peak vertical ground reaction force (vGRF), frontal plane knee valgus angle and moment, and increased sagittal plane knee extension from initial contact to 70ms post landing compared to the standard arrow condition.

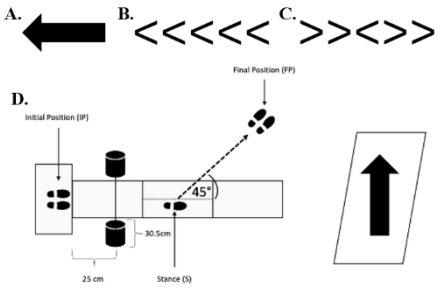

Fig 1: A. Standard arrow, B. congruent flanker, and C. incongruent flanker visual cues to change direction to left. D. Athletes jumped over a hurdle, landed unilaterally and cut towards the direction the visual cue.

Methods: 9 Division 1 athletes (Age: 19.82±1.67) completed a of series anticipated and unanticipated jumping tasks in response to both standard arrow, congruent, and incongruent flanker tasks to the left and right (Figure 1), though only the unanticipated standard arrow and incongruent flanker tasks were analysed in the present study (2 tasks total). Athletes were instructed to jump bilaterally from their initial starting position, land unilaterally and immediately follow the prompt to change direction at a 45° angle as far as possible. Data was collected until each participant completed 3 valid trials in each condition. Images appeared on the screen at the apex of the initial jump, triggered when the vGRF fell below 20 N. 3D marker-based motion capture (Qualisys) was synchronized with force platforms to quantify vGRF, frontal plane knee valgus angle and moment, and sagittal plane knee extension. The peaks were taken from initial contact to 70ms post landing. Differences in Peak vGRF, frontal plane knee angle and moments, and knee extension angles were analyzed via paired t-tests between unanticipated standard arrow and incongruent flanker conditions ($\alpha = 0.05$). Cohen's *d* effect sizes are presented.

Results & Discussion: The type of visual cue did not significantly influence ACL injury risk factors during unanticipated change of direction tasks. Peak vGRF (Arrow: 4.05±0.62 N/kg; Flanker: 4.05±0.62 N/kg, p=0.845, d=0.03), frontal plane knee angles (Arrow: 3.84±4.62°; Flanker: 2.72±6.11°, p=0.329, d=-0.21), frontal plane knee moments (Arrow: 0.59±0.44 Nm; Flanker: 0.60±0.44 Nm, p=0.947, d=0.03) or sagittal plane knee flexion angle (Arrow: 11.18±3.65°; Flanker: 13.35±5.12°, p=0.122, d=0.49) did not differ. While limited by a small sample size, the present results suggest increasing visual task complexity does not influence ACL injury risk factors during unanticipated tasks. While increasing cognitive load during planned conditions can increase ACL risk factors [5], adding an additional cognitive challenge while reducing planning time appears to have a negligible effect. These findings imply reducing planning time may be sufficient to screen for ACL injury risk factors.

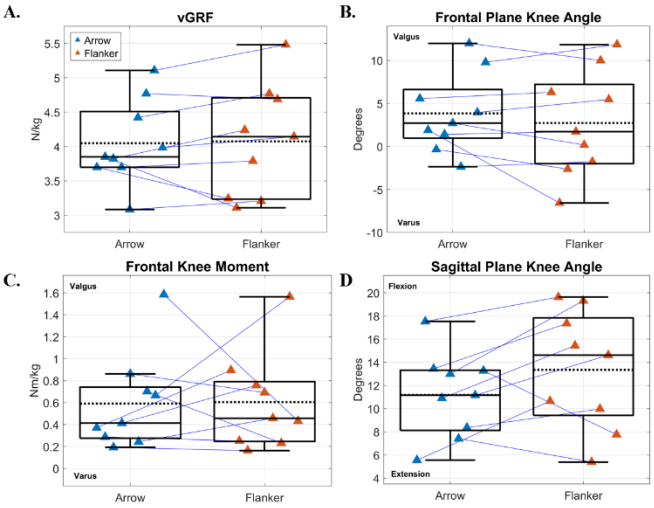

Fig 2: Boxplots and scatter plots for peak A. vGRF, frontal plane knee B. angles and C. moments and D. sagittal plane knee angles for each visual cue.

Significance: A standard unanticipated arrow task may be sufficient to screen for biomechanical risk factors associated with ACL injury, suggesting that increasing visual discriminatory task complexity may not be necessary.

References: [1] Almonroeder et al. (2018), *J Orthop Sports Phys Ther* 48(5); [2] Herman et al. (2016), *Am J Sports Med* 44(9); [3] Shibata et al. (2018), *Phys Ther Res* 21(2); [4] Bolt et al. (2024), *Sports Biomech* 23(10); [5] Hughes & Dai (2023), *Sports Biomech* 22(1); [6] Saadat et al. (2023), *Sports Biomech*